

Over-The-Shoulder Training Between Redundant Wearable Sensors for Unified Gesture Interactions

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ABSTRACT

Wearable computers are now prevalent, and it is not uncommon to see people wearing multiple wearable devices. These wearable devices are often equipped with sensors to detect the user's interactions and context. As more devices are worn on the user's body, there is an increasing redundancy between the sensors. For example, swiping gestures on a headphone are detected by its touch sensor, but the movement it caused can also be measured by the sensors in a smartwatch or smart rings. We present a new mechanism to train a gesture recognition model using redundant sensor data so that measurements from other sensors can be used to detect gestures performed on another device even if the device is missing. Our preliminary study with 13 participants revealed that a unified gesture recognition model for touch gestures achieved 89.2% accuracy for 25 gestures (5 gestures $\times$ 5 scenarios) where gestures were trained by leveraging the available sensors.

CCS CONCEPTS

• Human-centered computing $\rightarrow$ Gestural input.

KEYWORDS

Over-the-shoulder training, Unified Gesture Interactions, Wearables

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1 INTRODUCTION

The ever-increasing proliferation of smart devices has allowed people to incorporate technology into their lifestyles in the form of wearables [3]. Wearables such as glasses, headphones, rings, watches, and pendants are now equipped with processors and sensors and can communicate with other devices. These devices often act as an on-body interface, sensing interactions and behaviors, and/or providing output. Studies have shown that these wearable devices can enable quick microinteractions [2, 11, 13–15], more

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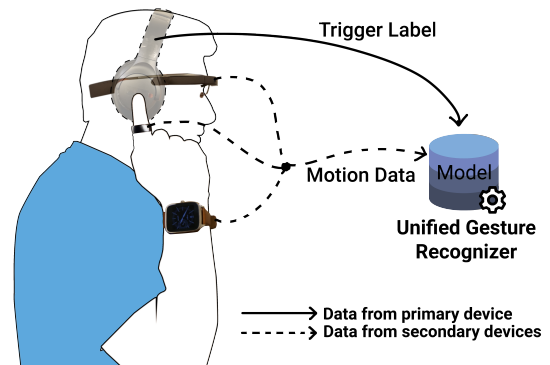


Figure 1: The concept of over-the-shoulder training. When a user uses gestures on a smart device (primary device), data from other devices (secondary devices) will also be collected to train a unified gesture recognizer model. The gesture recognized by the primary device will be used as a label for the training sample.

accessible interactions [1, 4, 5, 7–9, 17, 18], and even eyes-free interactions [12, 16, 20], by leveraging on its location on the user's body. However, these gestures are highly dependent on each device, which could cause frustration when the user wears a non-smart device or forgets to wear a smart device. In such scenarios, the user should find alternate ways for interaction, which may not be efficient.

In this work, we propose utilizing the input gesture recognized on a wearable device (primary) and the sensor data from all other connected wearable devices to train a unified gesture model to enable interactions for the primary device even when the device is absent. Wearable devices are equipped with various sensors such as inertial measurement units (IMU), microphones, and touch sensors. While worn on different locations on the body, their sensing is not completely exclusive and is often relatable to each other. For example, tapping the headphone's touchpad involves the movement of the tapping hand, which can be measured by the smartwatch's IMU on the wrist, and the collision between the hand and the headphone, which can be measured by the IMUs on smartwatch, smart glasses, and the headphones. As many smart wearable devices connect to a central device such as a smartphone, we believe that the data from those sensors can be collected together along with the label created by the intended interaction device to train a unified gesture recognition model (Figure 1). We named this process *over-the-shoulder training*.

The idea of exploiting relatable actions is not new. Researchers have demonstrated the feasibility of indirect sensing of gestures

using body-worn or in-pocket devices. For example, Scott et al. used a mobile device in the pocket to detect foot gestures [18] and Xu et al. used earbuds to detect touch gestures on face [19], without attaching a sensor or a device at the location of interaction. These methods use a gesture recognizer designed to indirectly detect a predefined set of gestures at a certain configuration that require an explicit training process. In contrast, our method does not need an explicit training or labeling process, and instead, uses the primary device input to label the sensor data during use. As a result, the gesture recognizer is trained automatically as the user performs gestures on their wearable devices over time. Once the model is sufficiently trained, our system allows the user to use gestures associated with the primary interaction device when it is absent.

To demonstrate the method's feasibility, we developed a machine-learning based unified gesture recognizer system that collects data from multiple body-worn devices. The system collects movement data from IMUs on a smartwatch, smart ring, and smart glasses, while the user performs touch gestures on a smartwatch, smart glasses, and headphones. Our preliminary study with 13 participants showed that, for 25 classes (5 touch gestures in 5 different device configurations), the recognizer model gave a classification accuracy of 93% without the primary interaction device. The study also showed that the model could achieve a gesture detection accuracy of 89.2% by re-training the generalized gesture recognizer model with only 5 additional gesture samples from a new user.

2 OVER-THE-SHOULDER TRAINING

We define *primary device* as the device that the user directly interacts with. We assume that the primary device can accurately and reliably recognize gestures so that we can use the input on the primary device to label sensor data. We define *secondary device* as all devices with an IMU that the user does not directly interact with.

We implemented a gesture classification model using Rocket [6] classifier, one of the state-of-the-art classifiers for efficient time-series classification in recent time [6]. Using Rocket, we transformed time series into features using random convolutional kernels and passing the features to a linear classifier. We used Ridge Regression [10] classifier for gesture classification. For data collection, we assumed a scenario where a user might wear the following smart devices – a smartwatch, a smart ring, smart headphones, and smart glasses. We simulated a smartwatch using a 9-DOF IMU (Sparkfun BNO080) and an Ergo Touchpad (28 × 33 mm), a smart ring using a 9-DOF IMU, and smart headphones by attaching an Ergo Touchpad (46 × 64 mm) to Soundcore Life Q20 headphones. For smart glasses, we used a Google Glass XE. When the system detects a gesture event from a primary device, it segments movement data from the IMUs on other devices for 1200 ms, starting from 700 ms before the gesture was detected by the primary device. The gesture segmentation window contains more movement data before the gesture event than after because touch events are triggered after the completion of the gesture.

3 PRELIMINARY STUDY

We conducted a preliminary study to investigate if measurements from secondary devices could be used to recognize primary device gestures without the primary device, and to test how many samples

were needed to train a personalized model. For this study, we prepared five configurations: C1) Primary (P): smart glasses, Secondary (S): smart ring and smartwatch; C2) P: smart headphones, S: smart ring and smart glasses; C3) P: smartwatch, S: smart ring and smart glasses; C4) P: smart headphones, S: smart ring and smart glasses; C5) P: smart glasses, S: smartwatch and smart ring. The smart ring was worn on the user's right index finger, and the smartwatch was worn on the user's left wrist (C1 - C4). For C5, the hand-positions were swapped. We chose five touch gestures for primary device interaction, which are four directional swipes and a tap. For primary device measurements, only the directional gesture label was collected. For secondary device measurements, the 3-axis acceleration and orientation of the device were collected at a sampling rate of 100 Hz.

We recruited 13 participants (*Male* = 9, *Female* = 4, *Mean Age* = 30.2) from our university. All participants were right-handed. The participants wore the devices and performed touch gestures on the primary devices for each configuration one by one. They performed each gesture 25 times. Each gesture was performed from the resting position, and the participants had to move their hands back to the resting position after performing the gesture. There was a one-second break between gestures to prevent incomplete resting. In total, we collected 5 configurations × 13 participants × 5 touch gestures × 25 samples = 8,125. There was a 1-minute break between configurations. The study took about an hour for a participant to complete. The study was approved by UVA IRB (#4652).

Using a within-subject training and testing (70 – 30 train-test split), our model achieved an accuracy of 93% and an F1 score of 92.7% for the 25 classes (5 gestures × 5 configurations). However, in a realistic scenario, the model may not have any of the new user's data. To test the performance of a generalized model trained without incorporating the target user's data, we trained our model using leave-one-participant-out (LOPO) cross-validation (between-subjects). The model showed 75% of accuracy and 73.2% of F1 score. To investigate how the model would improve over time as the user continues to use the system, we took the model without the new user's data and tested its performance while increasing the number of gesture samples from 2 to 16. Figure 2 shows the accuracy and the F1 score by the number of additional samples. The model trained with 6 additional samples could achieve over 90.5% of accuracy and 90.1% of F1 score.

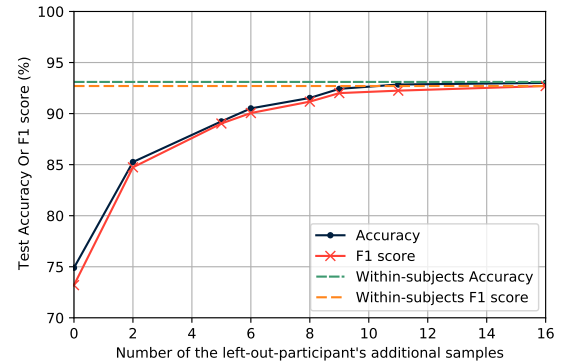


Figure 2: Results with the LOPO data plus the ignored participant's additional samples range from 2 to 16.

4 DISCUSSION AND FUTURE WORK

Our preliminary study showed that over-the-shoulder training method could enable an automatic training of a unified gesture recognizer and the trained recognizer could recognize primary device gestures with high accuracy, only using secondary device data. As people wear more and more wearable devices, we believe that this method will soon allow users to use familiar gestures that they use on certain smart devices even if they do not wear the exact device.

However, the study has several limitations that need to be addressed in the future work. First, people have different wearable devices and a generalized model may not exist. We did not test how many samples would be needed to obtain sufficient recognition performance. Second, the study only investigated the gesture classification performance and did not consider false activations, which can be crucial in actual deployment.

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